Big and small black holes follow the same rule

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Black holes come in very different sizes. Some weigh about 10 times as much as the Sun. Others weigh millions of times more. Scientists now say both kinds follow the same rule when they shoot **jets** of **material** into space.

The study was led by Dr. Adelle Goodwin in Australia and Dr. Andrew Mummery in Princeton.

The team studied times when a giant black hole pulls a star apart. Sometimes a jet shoots out right away. Other times, the jet appears months or even years later. Why?

The scientists found that late **jets** appear when the black hole's "eating" slows down to a certain point. Smaller **black holes** make **jets** at that same point.

Not everything near a black hole falls in. Some **material** blasts back out. Goodwin called it "a black hole burp."

The rule could help scientists **predict** when these **outbursts** will happen.

Word Treasure

black holes -- Black holes are places in space where gravity is so strong that nothing, not even light, can get out.

jets -- Jets are fast, narrow streams of material shooting out into space.

material -- Material is the stuff that things are made of, like gas and dust in space.

predict -- To predict means to say what will happen before it happens.

outbursts -- Outbursts are sudden, powerful blasts.

Background you need

A black hole has gravity so strong that nothing, not even light, can escape from it.

Some black holes are small, and others are giant ones found in the middle of galaxies.

When gas or pieces of a star swirl around a black hole, some of it can shoot back out into space as a jet.

Break it down

WHO: Scientists led by Dr. Adelle Goodwin in Australia and Dr. Andrew Mummery in Princeton
WHAT: Big and small black holes seem to follow the same rule for shooting jets
WHERE: Black holes far out in space, studied with telescopes
WHY: Scientists wanted to know why some jets appear right away and others years later

Why it matters

Finding one rule for black holes of all sizes helps us understand how the universe around us works.

Test what you remember

1. How much do some small black holes weigh?

- (A) Nothing at all
- (B) About as much as the Moon
- (C) About as much as Earth
- (D) About 10 times as much as the Sun

2. What do black holes shoot into space?

- (A) Light bulbs
- (B) Water
- (C) Jets of material
- (D) Rocks from Earth

3. What happens when a giant black hole pulls a star apart?

- (A) The Sun gets bigger
- (B) A jet may shoot out right away or much later
- (C) Nothing ever happens
- (D) A new planet appears

4. When do the late jets appear?

- (A) When the black hole's "eating" slows to a certain point
- (B) When the star grows back
- (C) Every 10 years
- (D) Only at night

5. What did Goodwin call the material blasting back out?

- (A) A black hole sneeze
- (B) A space cloud
- (C) A star cake
- (D) A black hole burp

6. How could this rule help scientists?

- (A) It could help them predict when outbursts happen
- (B) It could stop black holes
- (C) It could move stars
- (D) It could help them build black holes

Answer key (try the quiz first!): 1: D 2: C 3: B 4: A 5: D 6: A

Questions to think about

MY THOUGHTS (at least 20 words):